

Customer Shopping Behavior Analysis

Analyzing **3,900 purchase records** across product categories to uncover spending habits, customer segments, and strategic opportunities.

Dataset at a Glance

3,900

Purchase Records

18

Columns

37

Missing Values

Review Rating only

25

Unique Items

Features span demographics, purchase details, and shopping behavior ⁴ including subscription status, discounts, and review ratings.



Data Preparation in Python

Cleaned & Standardized

Snake_case columns; missing ratings filled by category median

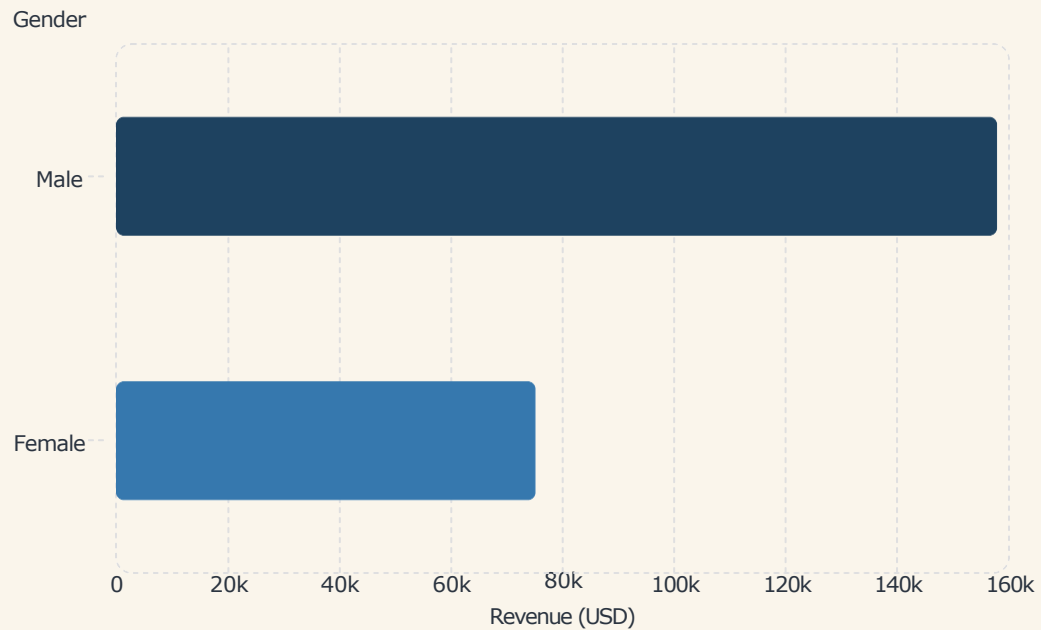
Feature Engineering

Created **age_group** and **purchase_frequency_days**

Database Ready

Loaded cleaned DataFrame into **PostgreSQL** for SQL analysis

Revenue by Gender



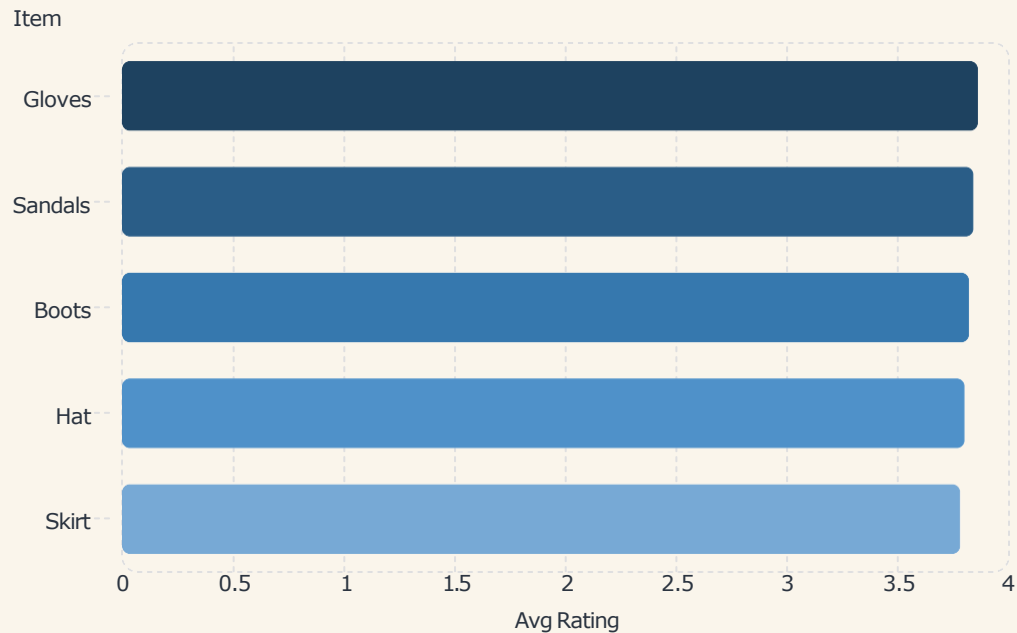
Key Finding

Male customers generate **21x more revenue** than female customers 4 a significant gap worth investigating.

Male: \$157,890 · Female: \$75,191

Product Ratings & Discount Sensitivity

Top 5 Products by Rating



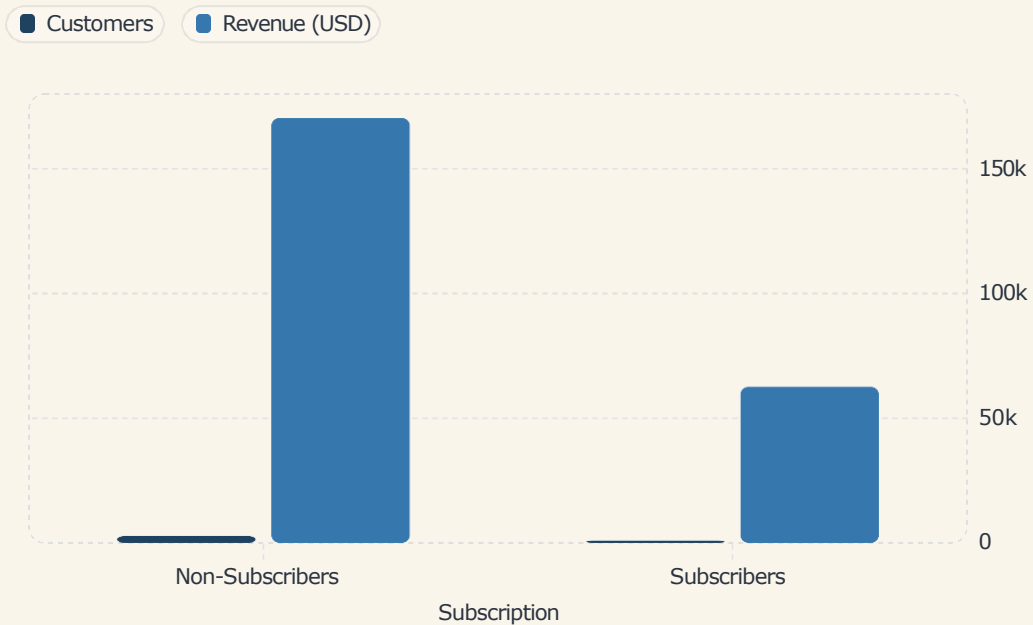
Most Discount-Dependent Products

Hat 4 50%

Sneakers 4 49.7%

Coat 4 49.1%

Subscribers vs. Non-Subscribers



Insight

Subscribers are **27%** of customers but spend nearly the same per transaction (~\$59.50 vs \$59.87).

i Opportunity: Grow subscriber base to lock in recurring revenue.



Customer Segmentation

Loyal

3,116 customers 4 the core base

Returning

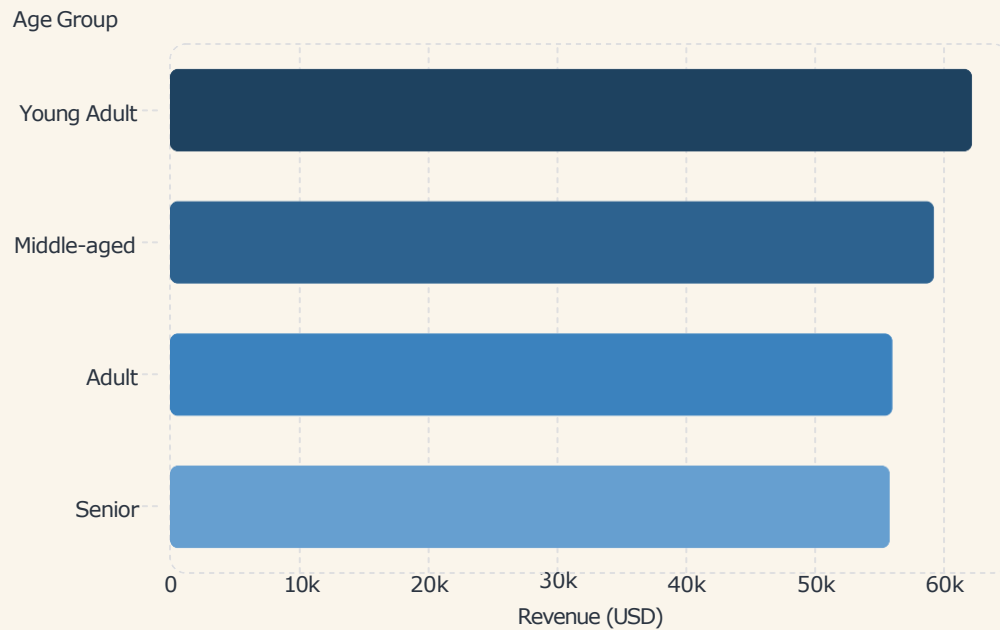
701 customers 4 growing potential

New

83 customers 4 early engagement needed

Repeat buyers with 5+ purchases: **958 subscribers** vs **2,518 non-subscribers**

Revenue by Age Group



Takeaway

Young Adults lead in revenue. All groups are relatively close & broad targeting is viable.

Express shipping avg: **\$60.48** vs Standard **\$58.46**

Business Recommendations

1	Boost Subscriptions Promote exclusive subscriber perks
2	Loyalty Programs Reward repeat buyers toward Loyal segment
3	Review Discount Policy Balance sales lifts with margins
4	Product Positioning Highlight top-rated, best-selling items
5	Targeted Marketing Focus on high-revenue age groups & Express shippers